

# CAD Viewer — User Manual

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**Version:** v0.81 · **URL:** <http://localhost:8088/>

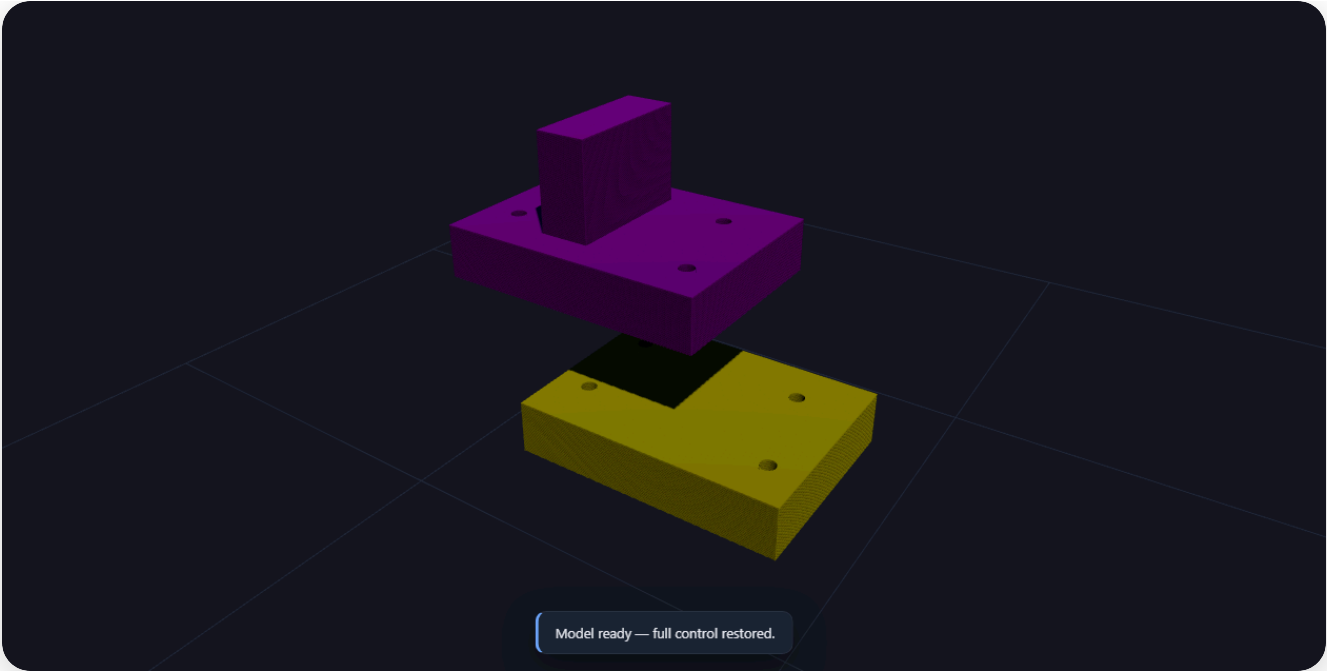
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## 1. Welcome — why CAD Viewer?

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**Turn a CAD file into a live, shared 3D scene.** CAD Viewer is a real-time collaborative browser app for viewing CAD geometry. Anyone on your network can open the model with nothing more than a web browser — no CAD software, no licences, no installs. Everyone shares the same view, and everyone can interact with it.

At a glance

Instead of emailing files back and forth, share a live model:

| Instead of...                         | CAD Viewer lets you...                                                               |
|---------------------------------------|--------------------------------------------------------------------------------------|
| Export → email → wait → open → repeat | Share one <b>join link</b> ; the model appears instantly                             |
| Sending multi-MB CAD bundles          | Convert to lightweight <b>GLB/GLTF</b> once and share a small file                   |
| Explaining with static screenshots    | <b>Point, move, measure and zoom together</b> on the live model                      |
| Guests installing CAD software        | Guests open the link in <b>any browser — nothing to install</b>                      |
| Worrying about files leaving your PC  | The model is <b>streamed to memory only</b> ; nothing is written to the guest's disk |

Application features

**► Both host and guest can interact.**

This is not a one-way broadcast — anyone in a session can orbit, zoom and pan, show or hide parts, highlight a part, move parts along X/Y/Z, measure distances, and chat — all mirrored live in both directions.

**► Live sync.**

Camera, part visibility, selection, part moves, measurements, transparency, the exploded view, and lighting/animation are shared between all members in real time. Late joiners automatically receive the current model and view state.

**► Measure & inspect.**

Click two corners of any part to read an exact distance in millimetres, and add notes in the session chat.

**► No install for guests.**

A guest needs only a browser — open a link and the shared model appears. Nothing to download, nothing to set up, no CAD licence.

**► Nothing is stored on the guest.**

The model is streamed to the guest's memory (RAM) only and rendered in the browser; no file is written to their disk. Close the tab and it's gone — ideal for sensitive geometry or NDA work.

**► Efficient sharing.**

STEP, IGES and STL files are converted to **GLB/GLTF** — a lightweight, web-native format — on the server, so one compact file moves fast across the network instead of a bulky CAD bundle.

**► Cross-platform & faithful.**

Works in any modern browser on Windows, macOS, Linux and tablets; part names, colours and materials survive the conversion, so reviewers see the design the way it was authored.

## What it helps with

**Online calibration of engineering design.** Because everyone shares the same live view and can interact with it, the design-review / calibration loop collapses from email-and-

export cycles into one shared session — questions are answered immediately, on the actual geometry, by pointing, measuring and looking together.

## Use cases

### ► Design review / sign-off

review the model together across offices or the shop floor instead of exchanging exports.

### ► Remote inspection

check a part or assembly before manufacturing, from anywhere on the network.

### ► Live design changes

everyone watches the part move as it happens.

### ► Client demo

show the design without installing CAD on the client's machine.

### ► Vendors & subcontractors

share the live model with a supplier to align on geometry, fit and tolerances before fabrication.

### ► Cross-site calibration

compare a real part against the shared model live, from a different location.

## 2. Starting the app

### Portable zip (Windows, no installs):

1. Unzip `cad-viewer-portable.zip` anywhere.
2. Double-click `start.bat` . It launches the server and opens the browser.
3. The app is served at **`http://localhost:8088/`**

**From source:** `npm start` (or `node src/server.js` ) in the project folder, then open the same URL.

### Changing the port

The default port is **8088**. Any of these overrides it:

| Method                                                                                                | Example                                                                                                              |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Launcher argument (zip)                                                                               | <code>start.bat 5555</code>                                                                                          |
| <code>port.txt</code> file next to the launcher<br>( <code>start.bat</code> / <code>start.sh</code> ) | create <code>port.txt</code> containing <code>5555</code>                                                            |
| Environment variable                                                                                  | <code>set PORT=5555</code> then <code>npm start</code> (Windows) /<br><code>PORT=5555 npm start</code> (macOS/Linux) |

Precedence: command-line argument → `port.txt` → env `PORT` → default 8088.

The join link and `/ip` address always reflect the actual port, so guests don't need to know it.

**STEP, IGES and STL files** (.step/.stp, .igs/.iges, .stl) additionally need

Docker + the `chair-cq:local` OpenCascade image. GLB/GLTF files work without any of it.

## Installing the STEP converter (Docker + OpenCascade)

To open **STEP / IGES / STL** files, the server needs a small converter built on the OpenCascade CAD kernel, which runs inside a **Docker** container. This is a **one-time** setup on the PC that runs the server — guests never need it.

Run `install-docker-opencascade.bat` (double-click it; it sits in the same folder as `start.bat` ). It walks through everything:

| Step                      | What the script does                                                                                                                                                                                                                                                                      |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Docker check</b>    | If Docker Desktop isn't installed, it installs it automatically via <code>winget</code> (~500 MB download) and starts it. You can also install it yourself from <a href="https://www.docker.com/products/docker-desktop/">https://www.docker.com/products/docker-desktop/</a> and re-run. |
| <b>2. Wait for engine</b> | Waits until the Docker engine is running (first start can take a few minutes).                                                                                                                                                                                                            |
| <b>3. Build the image</b> | Builds the <code>chair-cq:local</code> OpenCascade image from the bundled <code>Dockerfile</code> — first build downloads ~1 GB and can take <b>5–15 minutes</b> .                                                                                                                        |

| Step      | What the script does                                                                            |
|-----------|-------------------------------------------------------------------------------------------------|
| 4. Verify | Runs a quick check that the OpenCascade kernel is ready, so you know STEP conversion will work. |

**To install it manually** (instead of the script):

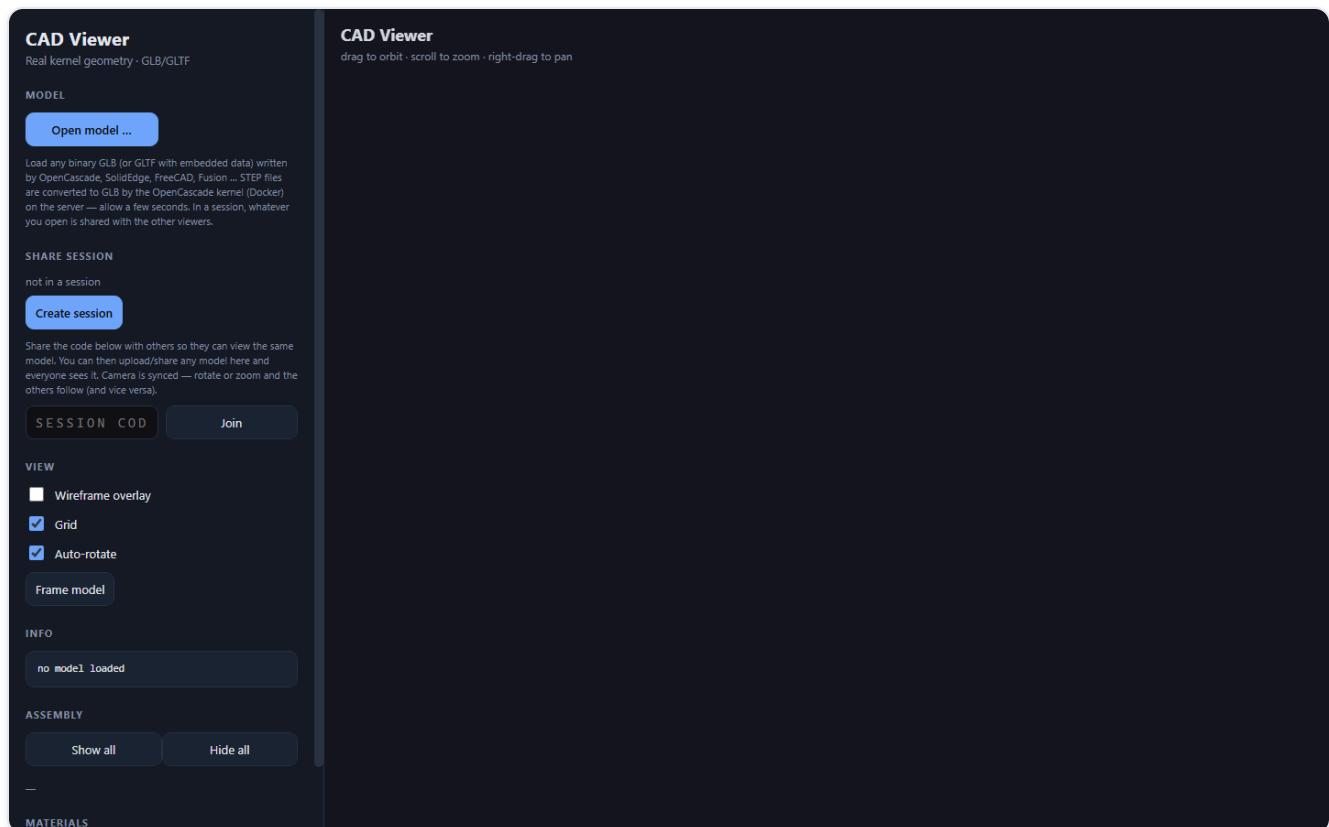
1. Install **Docker Desktop** and make sure the engine is running.
2. Open a terminal in the distribution folder and build the image:

```
```
```

```
docker build -t chair-cq:local .
```

```
```
```

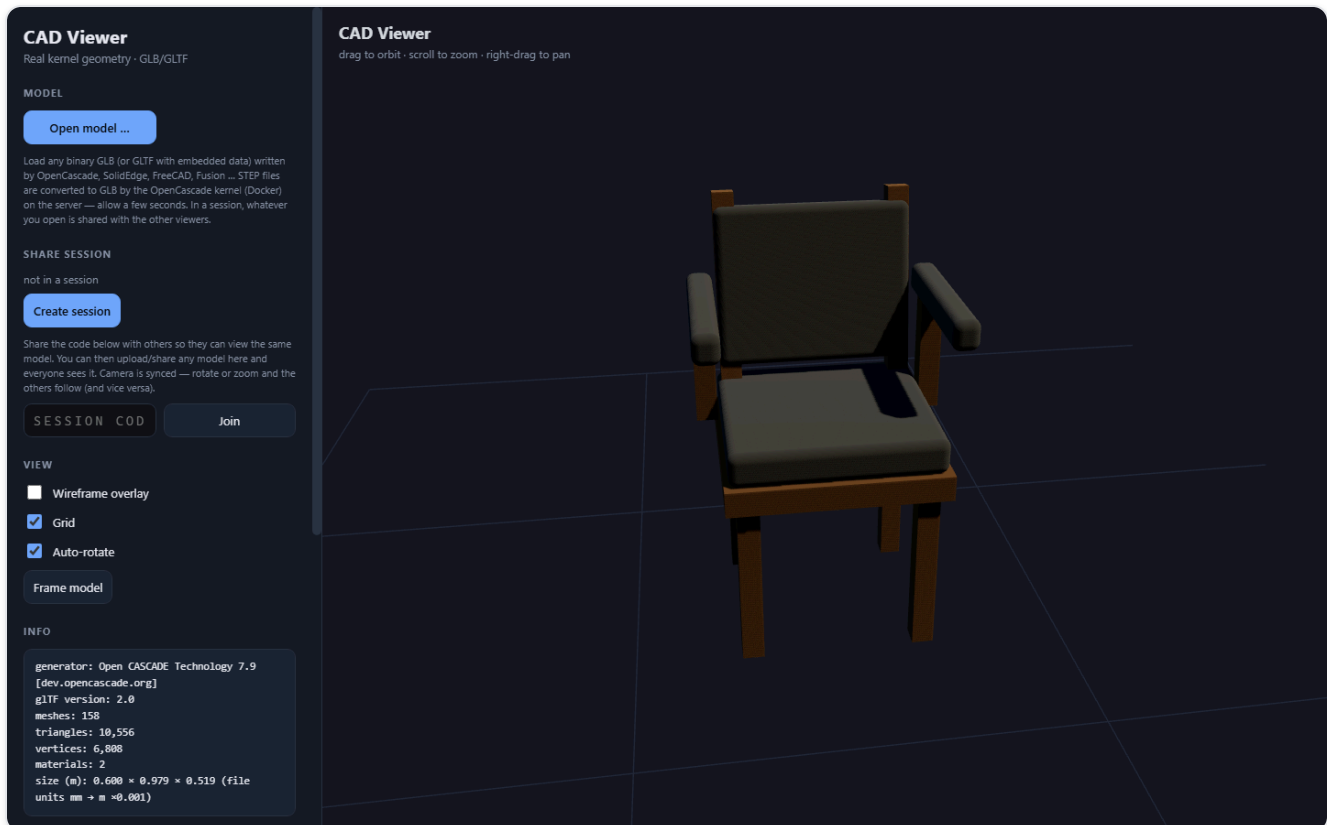
That's it. Once the image exists, STEP / IGES / STL files convert normally. If you only ever open **GLB / GLTF** files, you can skip this entirely.



## 3. Opening a model

1. Click **Open model...** in the left sidebar.
2. Choose a `.glb`, `.gltf`, `.step`, `.stp`, `.igs`, `.iges` or `.stl` file from your computer.

GLB/GLTF loads instantly. STEP, IGES and STL files are converted to GLB by the OpenCascade kernel on the server (a blocking overlay shows progress — a few seconds).



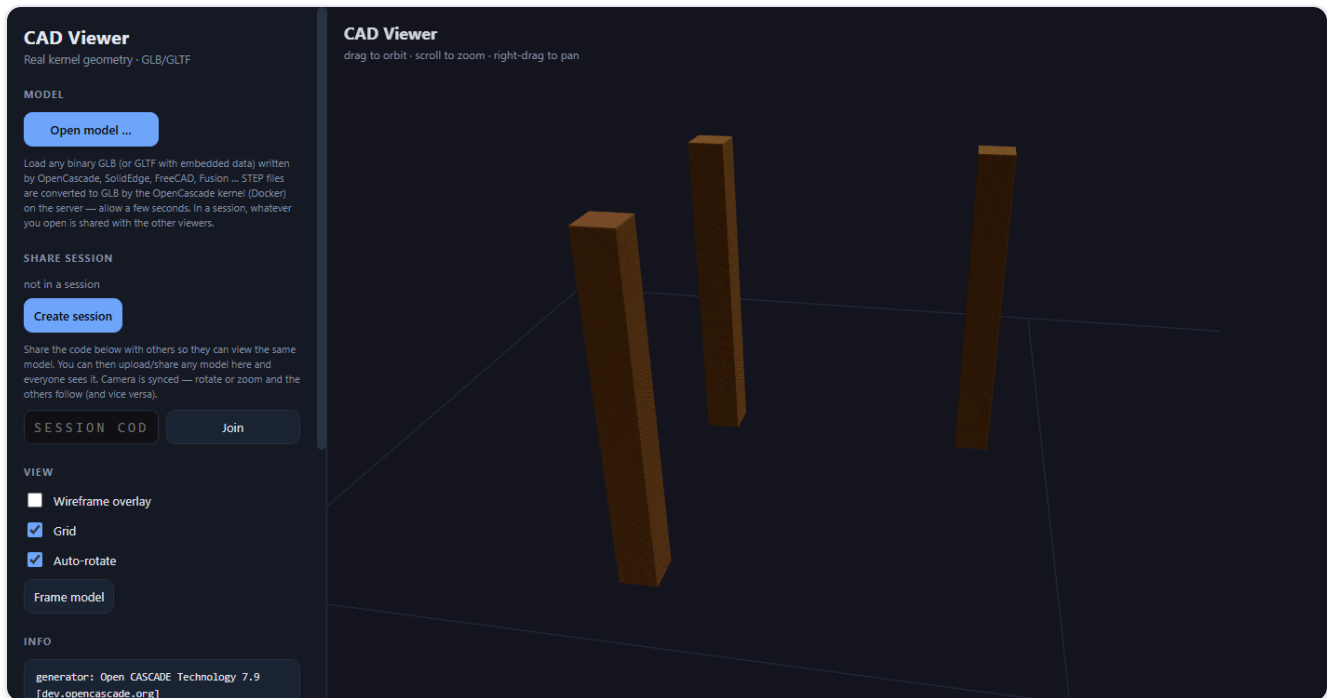
The sidebar shows you:

- **Info** — generator, mesh/triangle counts, dimensions and units
- **Assembly** — the part tree (see §4)
- **Materials** — swatches of every colour in the model

## 4. The Assembly panel — parts and visibility

Every part of the model appears in the **Assembly** tree. The tree opens expanded to **level 2** (the top assembly plus its direct children); deeper

levels are collapsed until you expand them.



## Expand / collapse

- Rows that contain sub-parts show a **+/- toggle** on the left.
- Click **+** to expand (show children), **-** to collapse (hide them).
- Expanding/collapsing is purely visual — it never changes part visibility.

## Show / hide parts

- **Checkbox** per part: uncheck to hide it in the viewport, check to show.
- Toggling a parent cascades to its whole subtree — turning a sub-assembly on

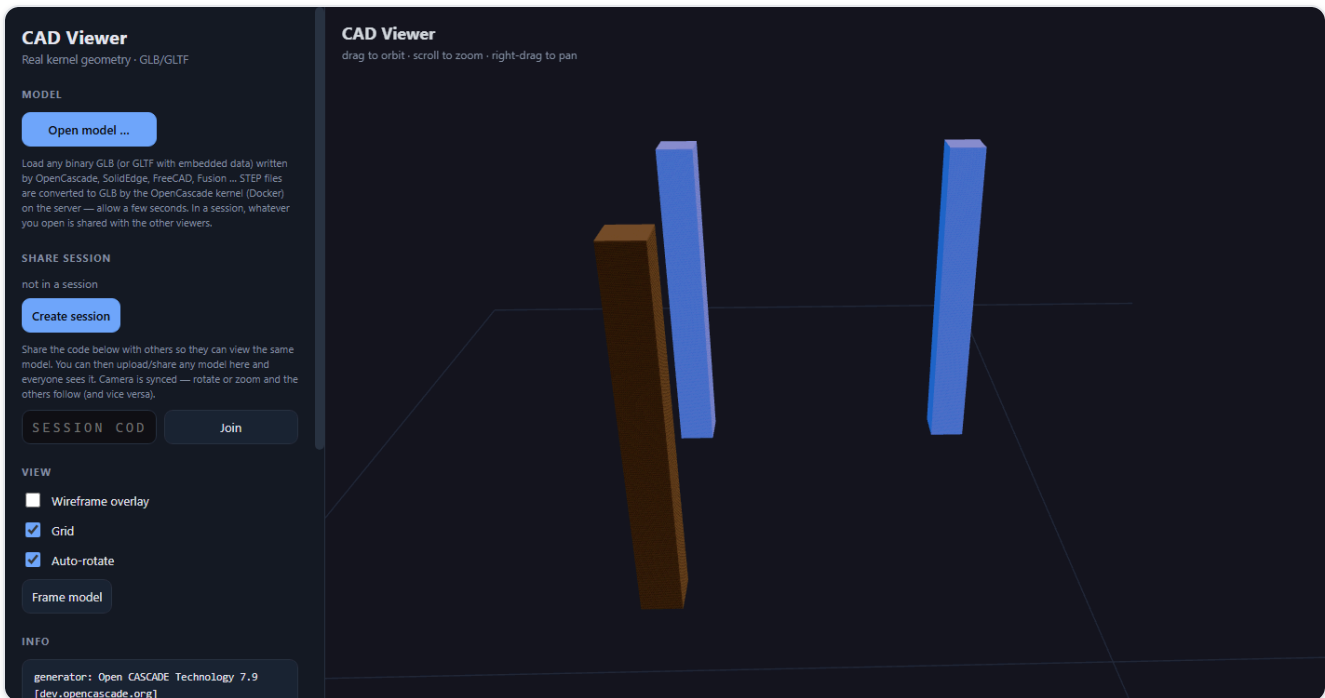
brings all its children back with it (three.js needs the whole chain visible to render anything).

- **Show all / Hide all** buttons apply to the whole model at once.

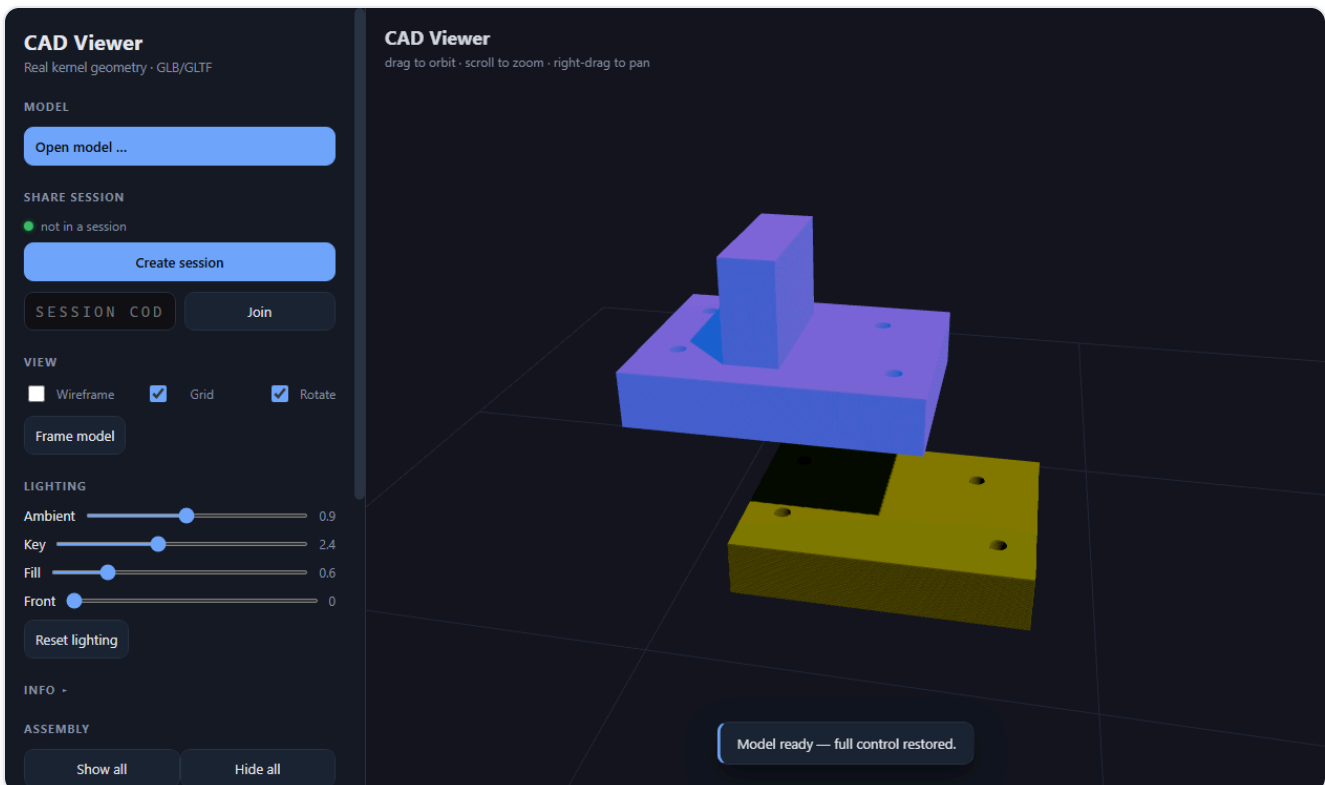
## Highlight a part (click the name, or click it in the viewport)

Click a part's **name** to highlight it in blue in the viewport — or **\*\*click the part directly in the 3D viewport\*\*** to select it the same way (left-click on the

part; left-click empty space to clear). Click another part to move the highlight; click the same part again to clear it. The highlight uses per-mesh material copies, so other parts sharing the same source material are unaffected.

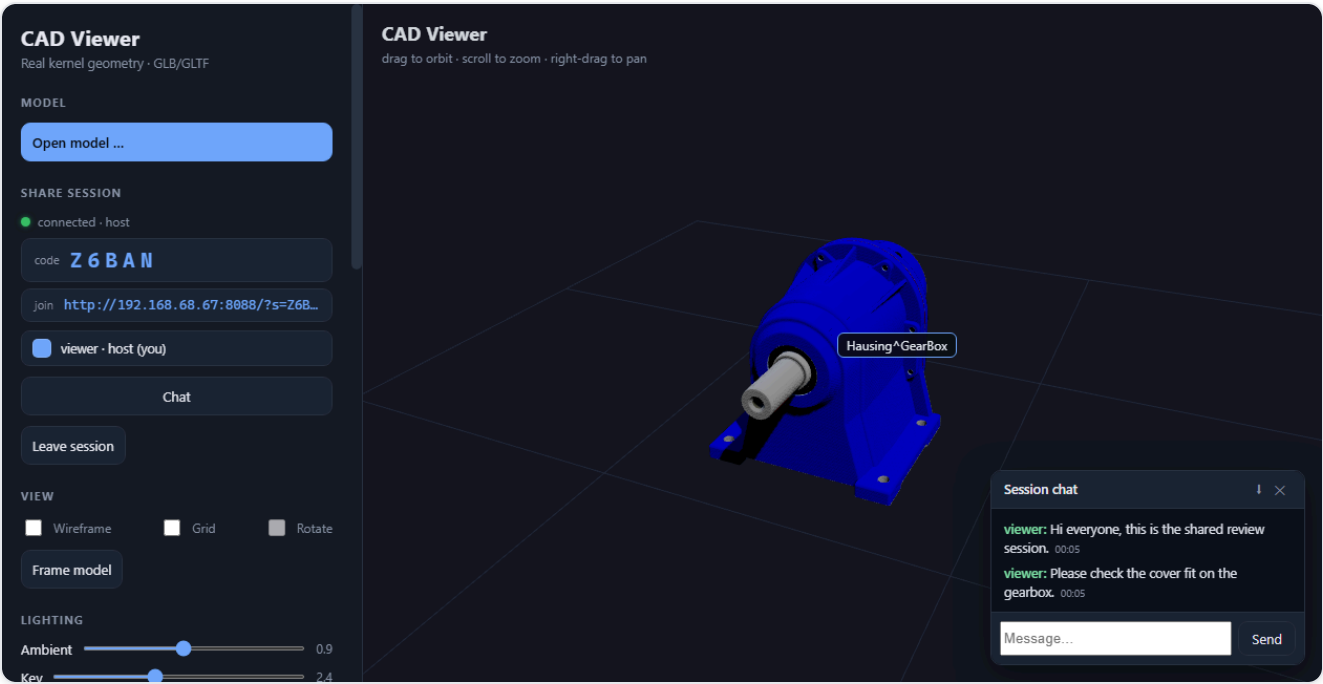


Clicking a part directly in the 3D viewport highlights it the same way:



Part name on hover

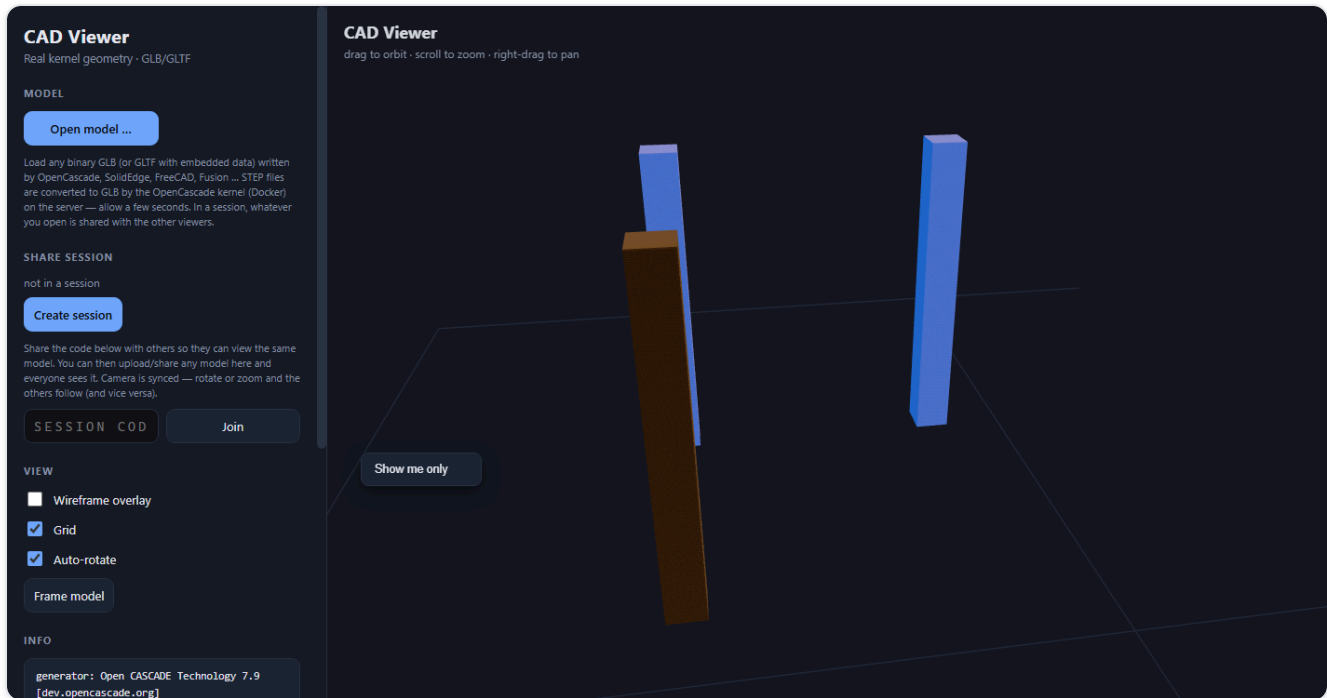
Move the mouse over a part in the 3D viewport and a small **tooltip** shows the part's name at the cursor, while its row in the Assembly tree highlights. This answers "which part is this?" instantly, without clicking.



Right-click menu (Hide / Move / Show me only / Make transparent)

Right-click a part — either its name in the tree, or the part directly in the viewport — to open a menu with four actions:

| Action           | What it does                                                                                                                               |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Hide part        | Turns that part (and its subtree) off in the viewport                                                                                      |
| Move part        | Selects the part and arms the axis gizmo so you can move it (see §5)                                                                       |
| Show me only     | Hides everything except that part and its children; ancestors stay visible so the isolated part still renders                              |
| Make transparent | Renders the part at 25% opacity so you can see through it (see §8); the item reads <b>Make opaque</b> when the part is already transparent |



## 5. Moving a part

A part can be moved along the X, Y or Z axis using the axis gizmo, so you can reposition a component of an assembly.

### Start a move

1. **Select the part** — click its name in the tree, or click it directly in the 3D viewport.
1. **Right-click** it and choose **Move part**. The X/Y/Z axis gizmo appears at the part.
1. **Click the gizmo arrow** for the direction you want to move (X = red, Y = green, Z = blue). That arrow glows to show it is armed.

### Drag to move

With an axis chosen, **left-drag** in the viewport to slide the part along that

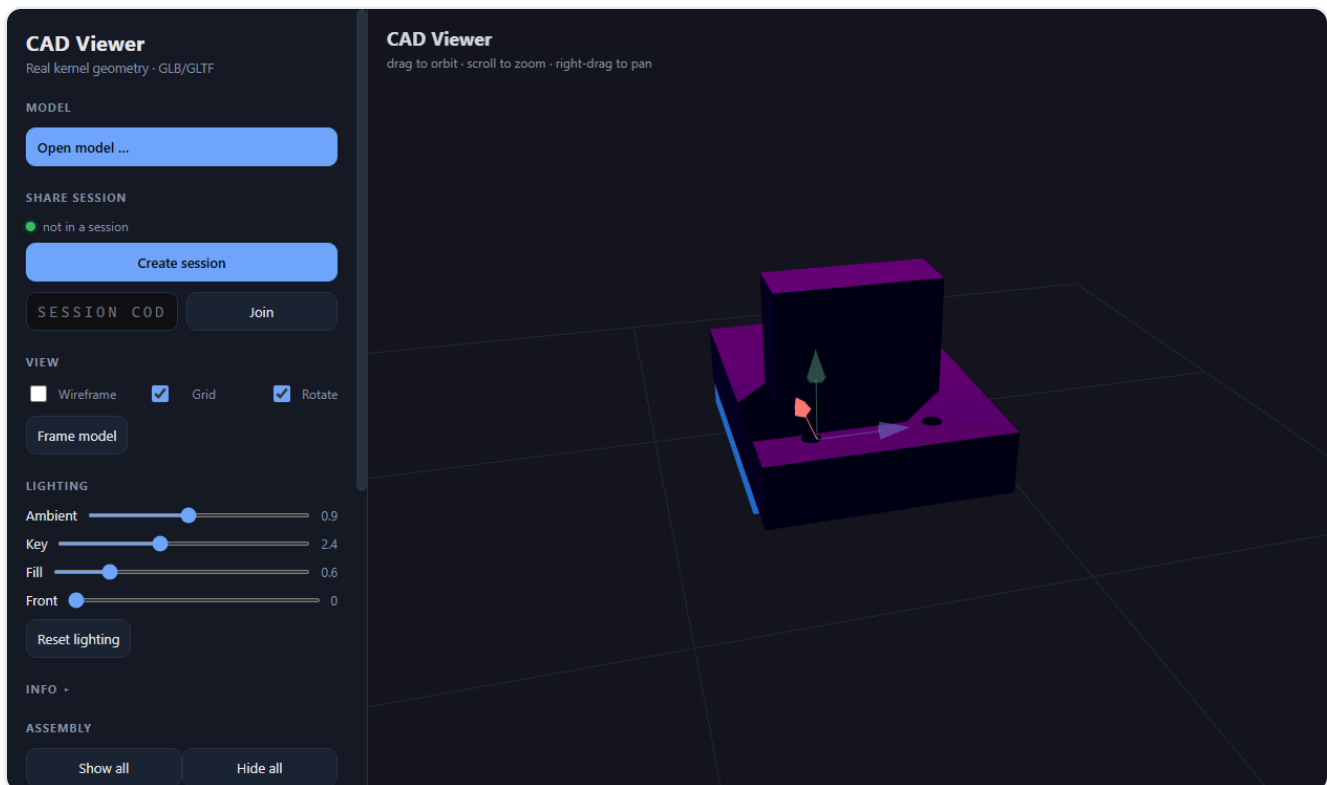
axis. The selected **part moves by itself** — its siblings stay put. (You can also pick an axis from the sidebar's **Move part** checkbox + **X / Y / Z** keys, or click a different gizmo arrow to change direction mid-move.)

## The gizmo

The X/Y/Z arrows (red/green/blue) stay about **1/8 of the screen** at any zoom level — zoom in and they shrink to keep that size. The armed arrow glows brighter than the others.

## Undo & reset

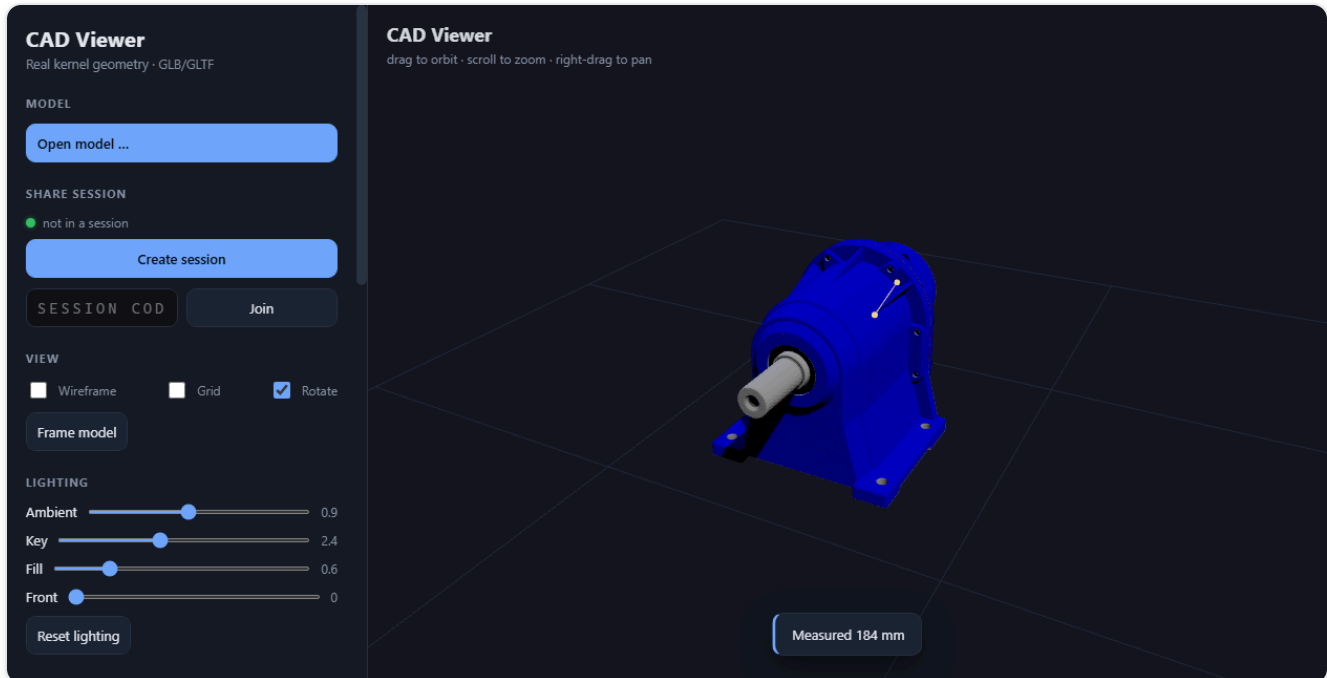
- **Undo** steps back through your last part movements (up to 200).
- **Reset** returns all parts to their original positions, collapses the exploded view, and clears the move history.



**In a session, part moves sync to every member** — move a part and the others watch it move live, and can move it back.

## 6. Measuring distances

The **Measure** tool reports the exact distance between two corners of a part in millimetres — handy for checking a dimension during review.



### How to measure

1. Tick the **Measure** checkbox in the Assembly panel (it turns **Move part** off — the two are mutually exclusive).

1. Move the mouse over a part. As you get near a **corner**, a small amber glow snaps to it and a live readout appears.

1. **Click the first corner**, then **click the second corner**. An amber dimension line is drawn between them and added to the measurements list.

### The measurements list

Each measurement is shown in the list with:

- the **distance** (mm) and its elevation angle,
- the **P1** and **P2** corner positions (mm),

- its azimuth angle.

Use the **X** next to an entry to delete it, or **Clear** to remove all. **Esc** cancels the in-progress first point.

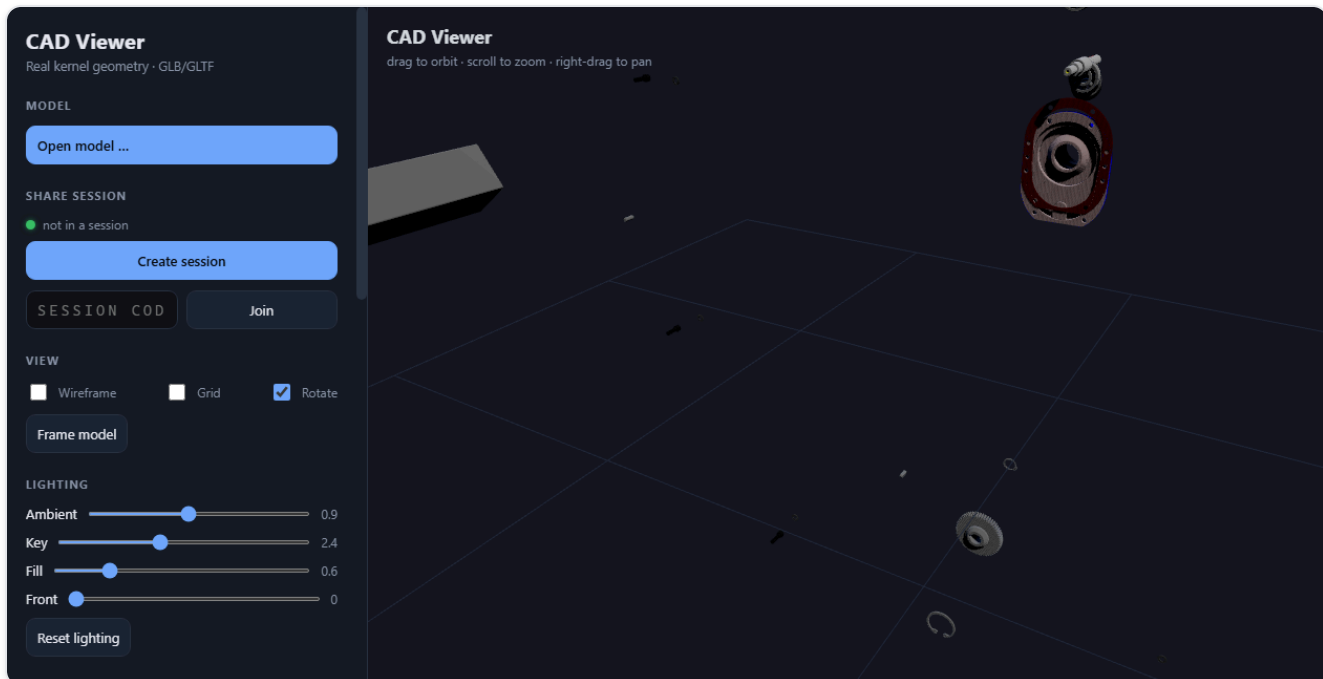
**Corners only.** The tool snaps to the part's corners (sharp edges), so it

gives meaningful points rather than random spots on a face.

**In a session, committed measurements sync to every member** — and late joiners see the measurements that were already made.

## 7. Exploded view

The **Explode** slider spreads the assembly's parts outward so you can see how it is put together — each part slides away from the assembly centre along its own direction.



## Controls

- **Explode** slider — drag from **0%** (assembled) to **100%** (spread apart).

- **Dir** — choose **Radial** (outward from the centre) or an explicit **\*\*X / Y /**

**Z\*\*** axis. Parts on the + side move +, on the – side move –.

- **Level** — **All parts** separates every individual part; **Top sub-assembly**

keeps each top-level sub-assembly together as a rigid unit.

The explosion is **non-destructive**: sliding back to 0% returns every part exactly to where it was (even parts you've moved). Pressing **Reset** in the Move panel also collapses the explosion.

**In a session, the explode state syncs to every member** — whoever moves the slider, everyone follows.

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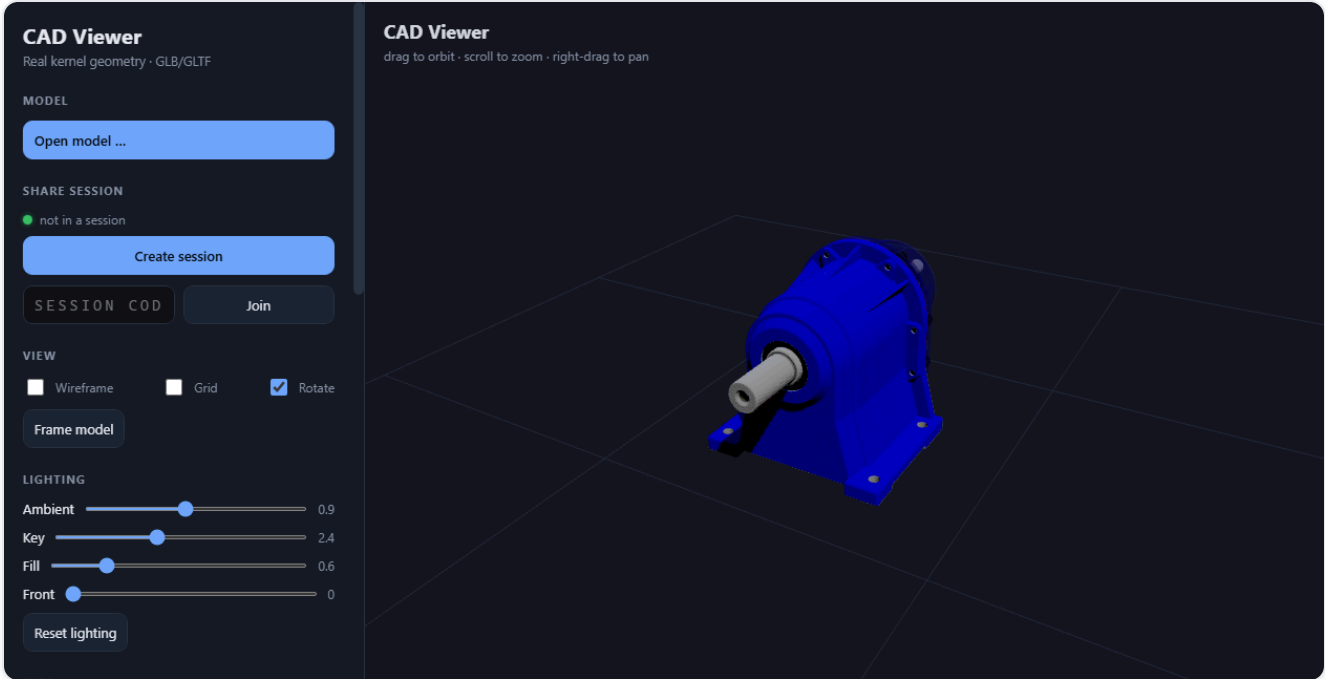
## 8. Part transparency

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Make a part **transparent** so you can see through it to the geometry behind — great for inspecting a housing, cover, or how parts nest.

1. **Right-click** a part (its name in the tree, or the part in the viewport).
2. Choose **Make transparent**. The part renders at 25% opacity.
3. To restore it, right-click again and choose **Make opaque** (the menu item

reflects the current state).



Transparency composes cleanly with selection — you can select, move, and measure a transparent part without it snapping back to opaque. \*\*In a session, transparency syncs to every member.\*\*

## 9. Navigating the viewport

| Action                    | Effect                                      |
|---------------------------|---------------------------------------------|
| <b>Drag</b> (left button) | Orbit the camera                            |
| <b>Scroll</b>             | Zoom                                        |
| <b>Right-drag</b>         | Pan                                         |
| <b>Frame model</b> button | Reset the view to fit the model             |
| <b>Wireframe overlay</b>  | Toggle wireframe on all parts               |
| <b>Grid</b>               | Toggle the ground grid                      |
| <b>Auto-rotate</b>        | Slow turntable spin (disabled in a session) |

## 10. Lighting & animation

### Lighting

A **Lighting** section in the sidebar lets you tune the light on the model:

| Control        | What it does                                                                  | Range | Default |
|----------------|-------------------------------------------------------------------------------|-------|---------|
| <b>Ambient</b> | Even fill from the sky and ground                                             | 0–2   | 0.9     |
| <b>Key</b>     | Main directional light (casts shadows)                                        | 0–6   | 2.4     |
| <b>Fill</b>    | Cool fill light from the opposite side                                        | 0–3   | 0.6     |
| <b>Front</b>   | Light straight from the viewer's side — handy when the front face is too dark | 0–3   | 0 (off) |

Drag a slider and the model updates live. **Reset lighting** restores the defaults.

### Animation

If the GLB you load contains **keyframe animation** (e.g. from Blender or a game pipeline — CAD-converted STEP/IGES/STL files have none), an **Animation** section appears with:

| Control             | What it does                               |
|---------------------|--------------------------------------------|
| <b>Play / Pause</b> | Start or stop the clip                     |
| <b>Loop</b>         | Repeat the clip (off = play once and stop) |
| <b>Speed</b>        | Playback speed 0.25×–4×                    |
| <b>Clip</b>         | Pick a clip when the model has several     |

### Session sync

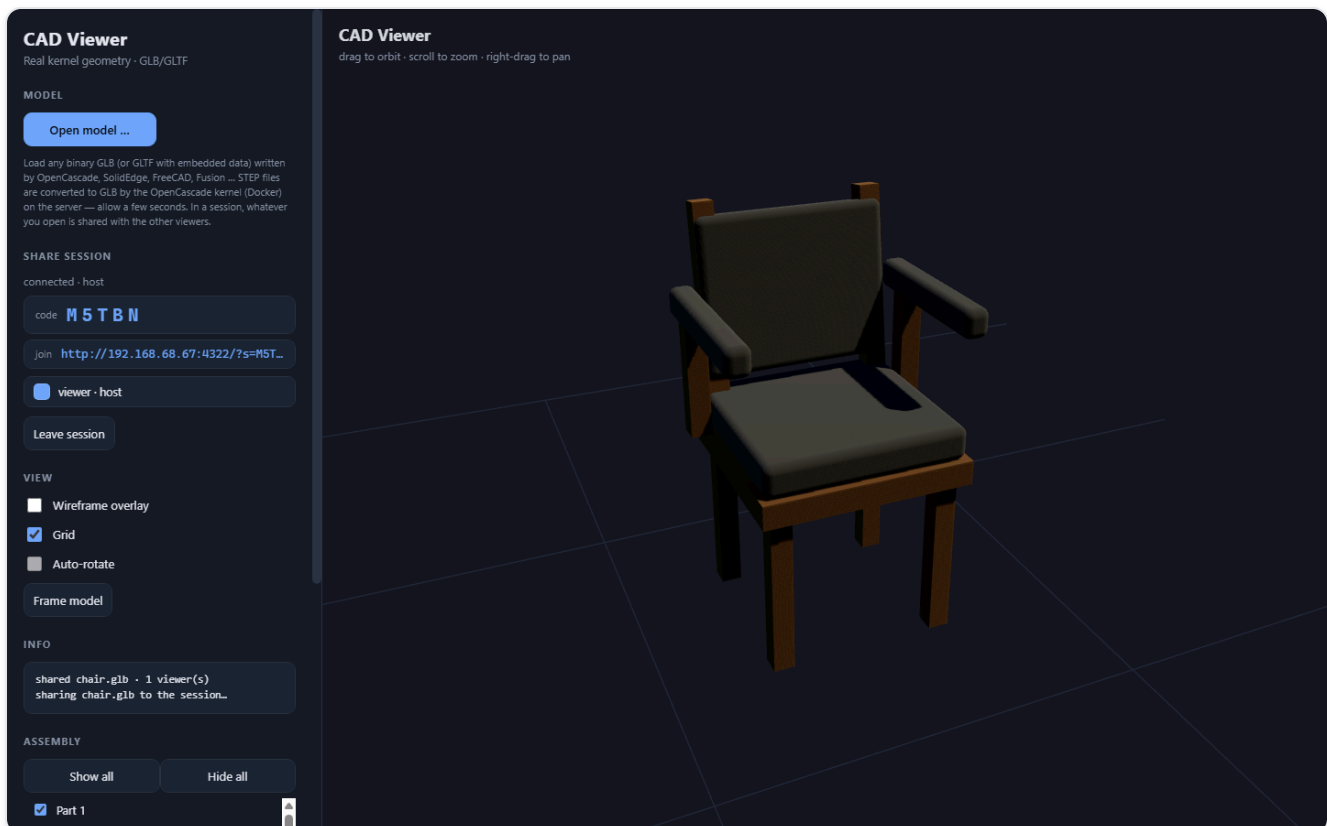
In a session, the **lighting levels (Ambient/Key/Fill/Front)** and the **animation state (clip, play/pause, loop, speed)** are shared with the other viewers, and late joiners receive the current settings. Camera, part visibility, tree expand/collapse, selection, part moves, measurements, transparency and the exploded view sync as well (see §12).

## 11. Collaborative sessions (host)

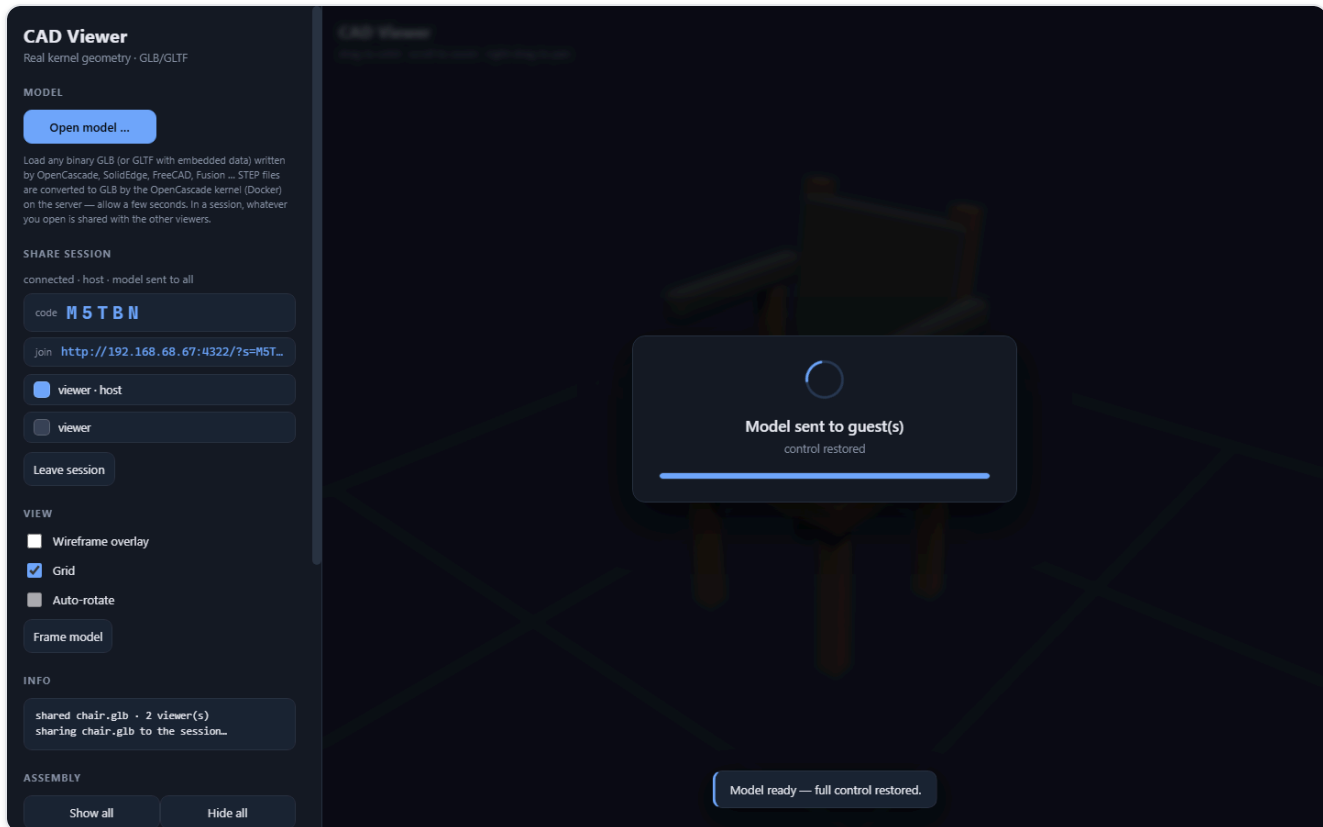
Sessions let other people on your LAN view the same model and follow your camera and part visibility.

On launch you're asked to **enter your name**; it is shown to the other viewers in the roster (e.g. Alice · host , Bob (you) ). Your name is remembered for next time.

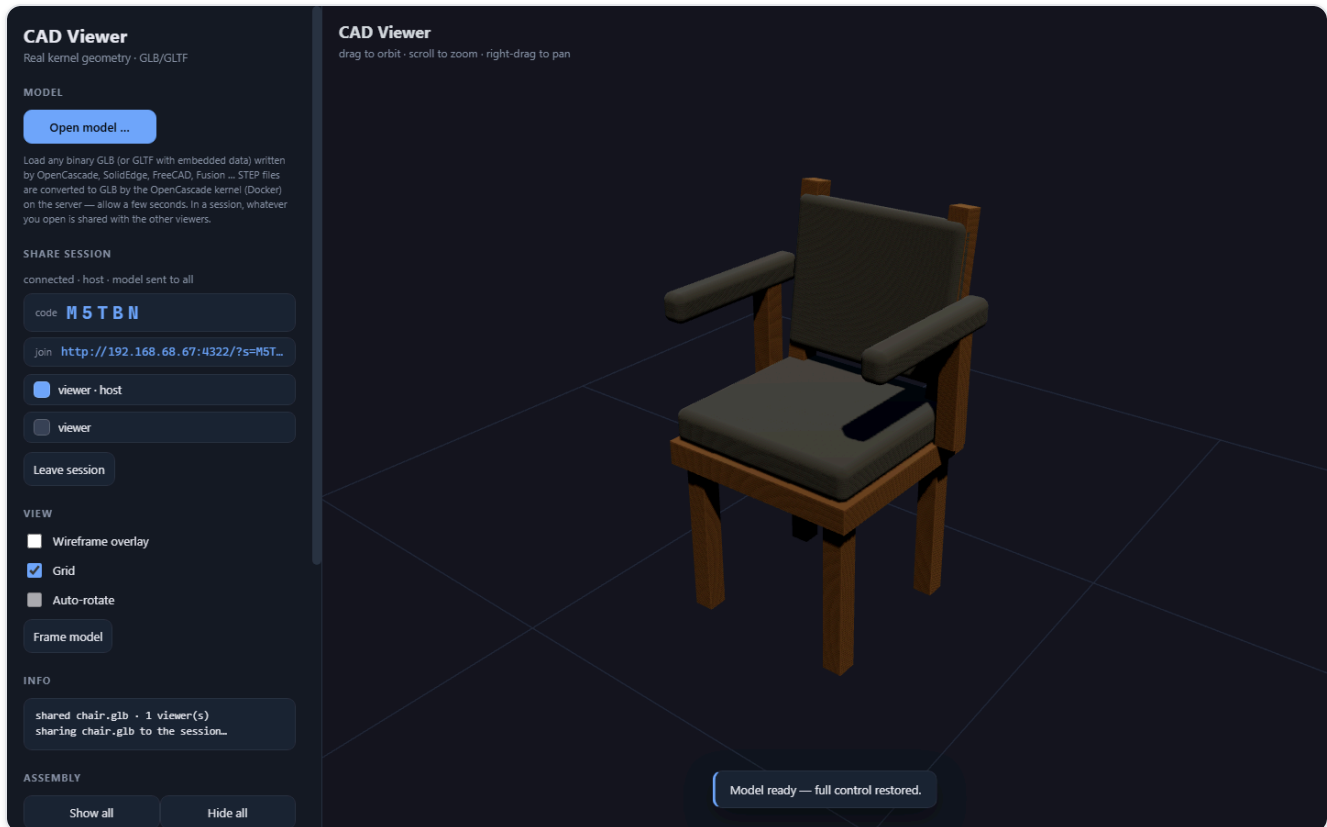
1. Click **Create session** — a 5-character code appears.
2. **Open a model** (or re-share one you've already loaded).
3. Share the **join link** shown under the code, or the code itself, with the other users.



While the model is being handed to guests you'll see a blocking **“Sending model to guest(s)...”** overlay; control is restored automatically once every connected guest confirms receipt.



The roster under the code shows who is connected. The **host** can remove a viewer by clicking **kick** next to their name; the removed viewer's screen is cleared and they see *"removed by host"*. A small status dot next to "Share session" shows whether the server is reachable (green = up, red = down).



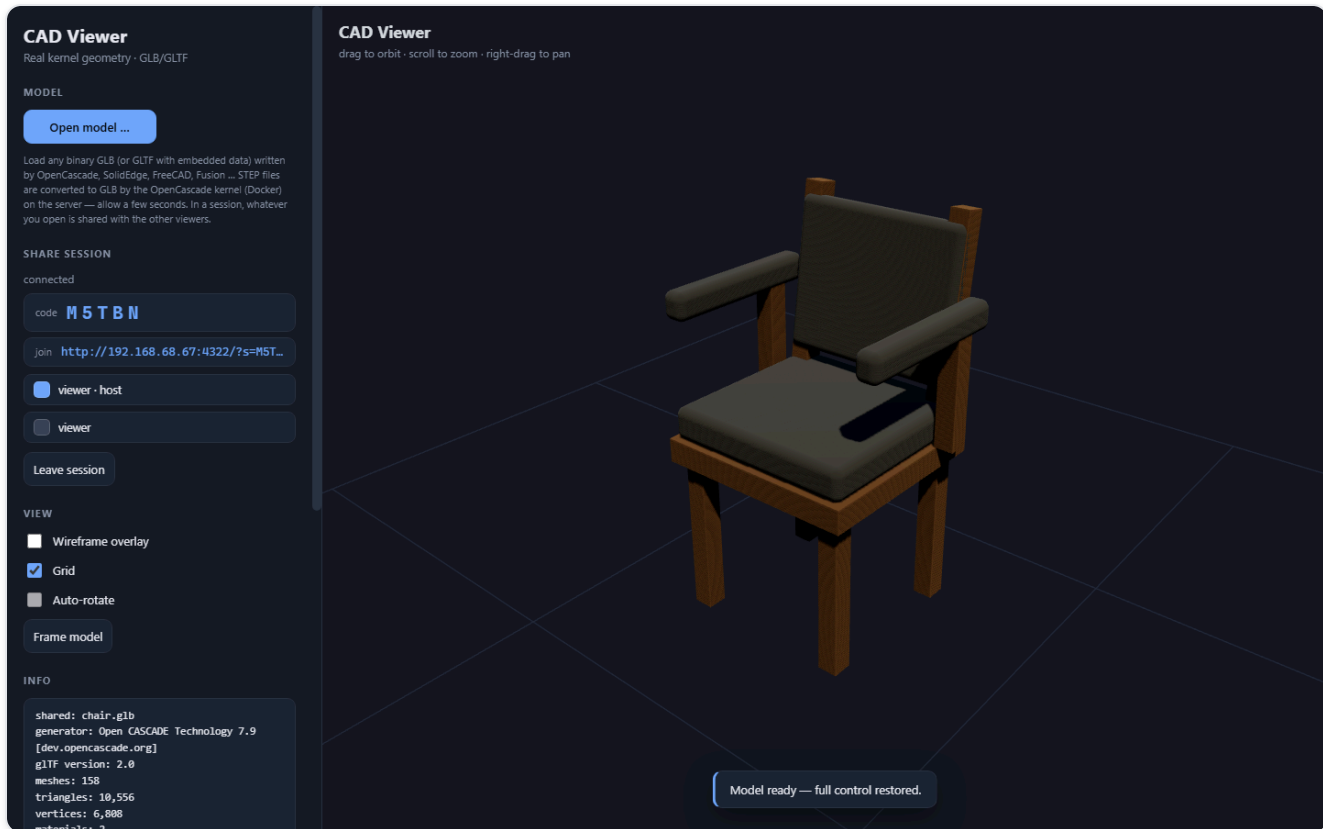
The **chat window opens automatically** when you create or join a session (see §13) — just start typing.

## 12. Joining a session (guest)

On another computer (same network):

1. Open the **join link** the host sent you, **or**
2. Open the app, type the code into **session code** and click **Join**.

You'll see a **“Receiving model...”** overlay with byte progress, then the model appears — same view, same parts, same visibility as the host.



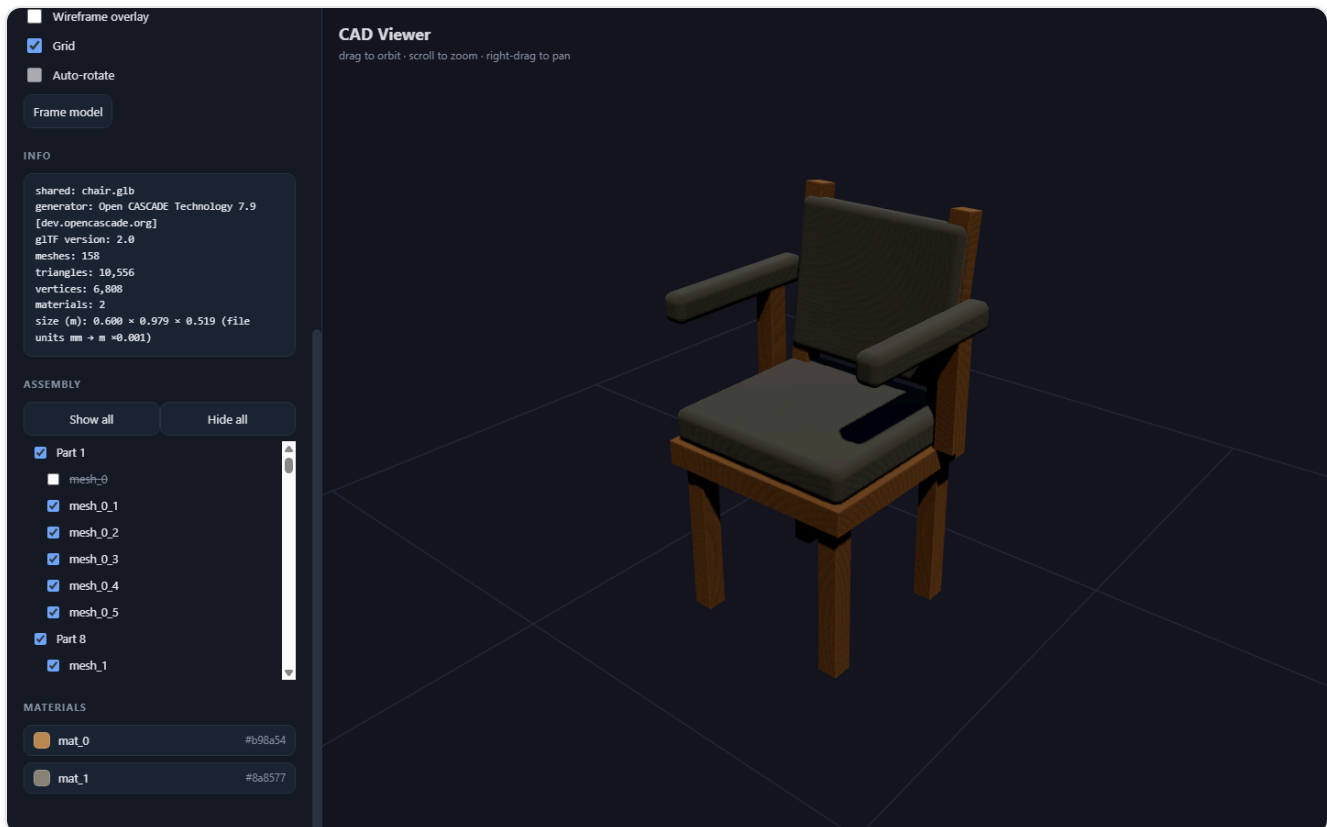
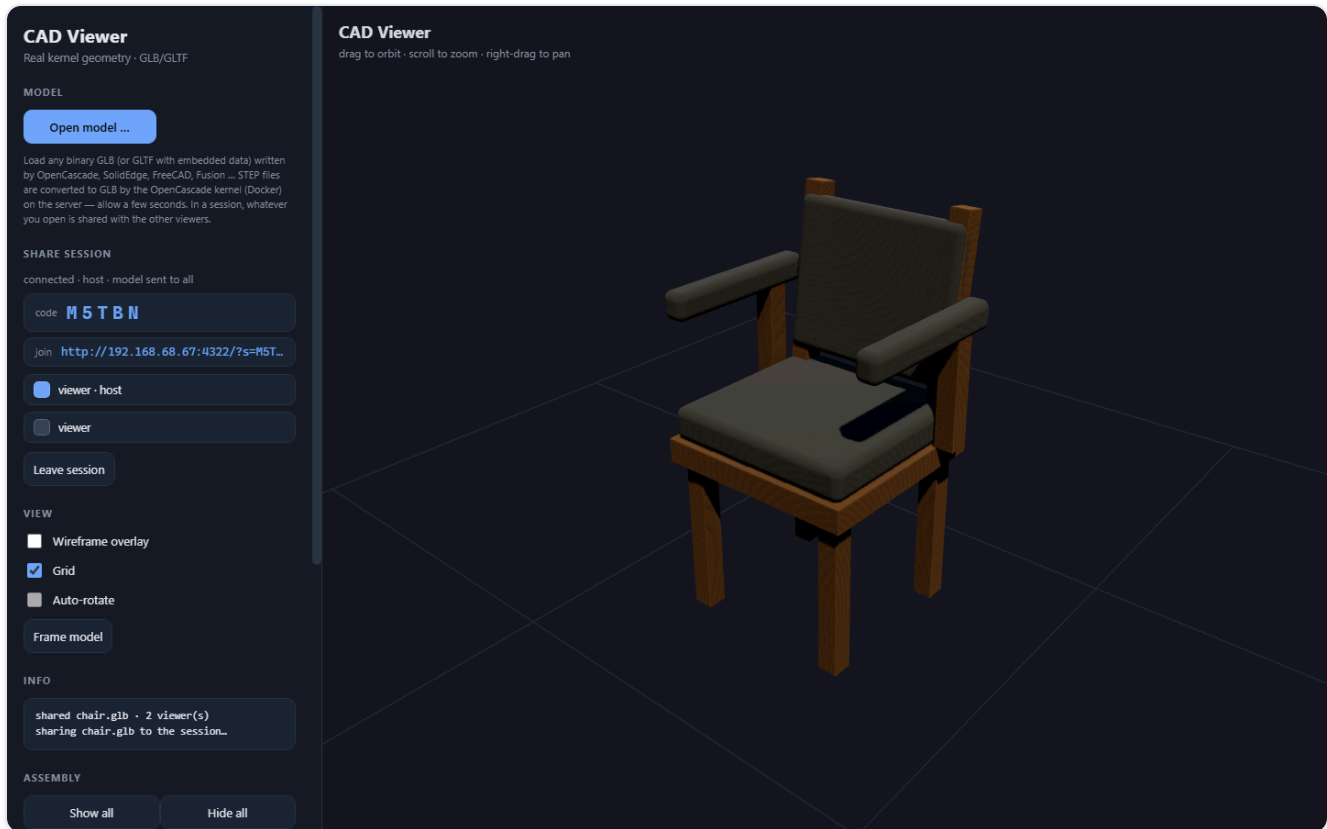
## What stays in sync

- **Camera** — whoever orbits/zooms/pans, everyone follows (both directions)
- **Part visibility** — show/hide any part and it changes for all members
- **Part moves** — moving a part updates it for everyone
- **Measurements** — a dimension you add appears for everyone
- **Transparency & explode** — part transparency and the exploded view match
- **The model itself** — late joiners automatically receive the current model

and visibility state

- **Expand/collapse & selection** — the assembly-tree view and part highlight

are mirrored live across members



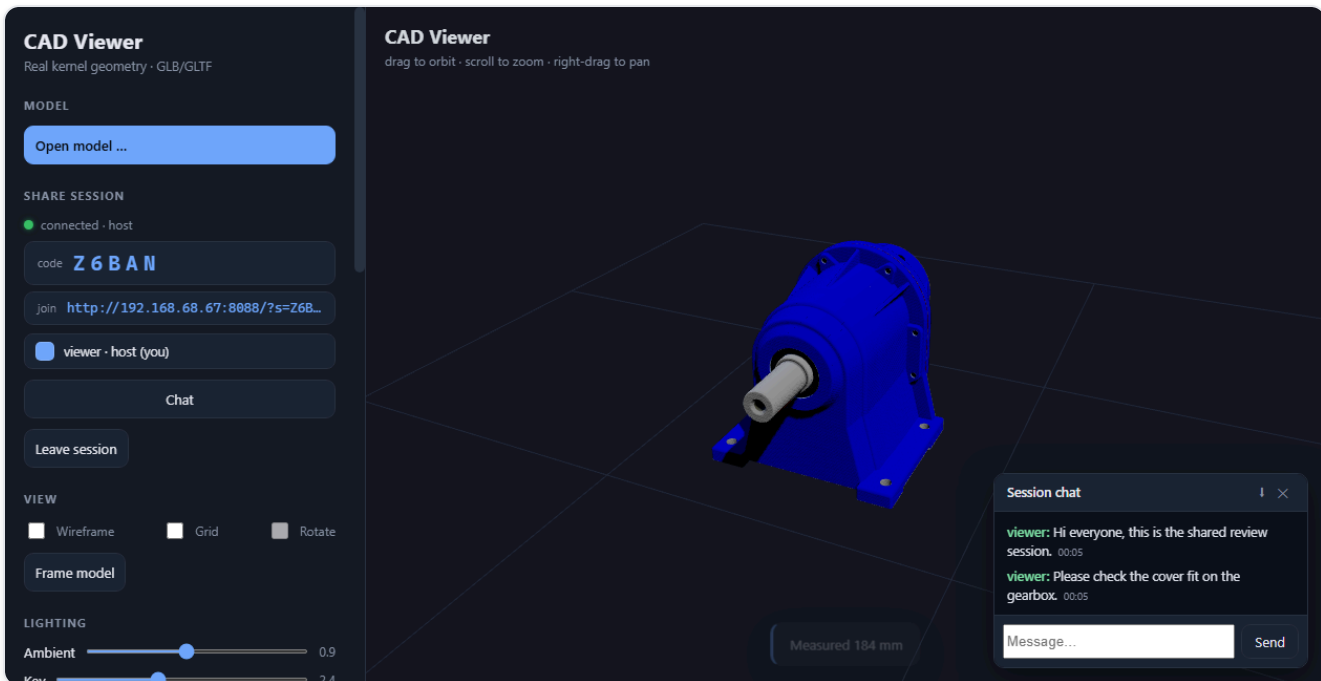
**Security — nothing is stored on the guest.** The model is \*\*streamed to the

guest's memory only **and rendered in the browser**. No file is written to

the guest's disk<sup>\*\*</sup>; closing the tab discards it.

## 13. Session chat

Every session has a built-in **chat window** so members can talk while they review. It opens automatically when you create or join a session.



### Chatting

- Type in the message box at the bottom and press **Enter** (or click **Send**).
- Each message shows the **sender's name**, the **text**, and the **time**; your

own messages are highlighted in green.

- Close the window with **X** and reopen it with the **Chat** button in the session panel.

### Chat history

- Messages are kept on the server (up to 200) and **replayed to late joiners**,

so a new member sees the whole conversation, not just what's said after they join.

- Click the ↓ button in the chat header to **download the transcript** as a

.txt file ( chat-<session-code>-<date>.txt ). Each line is

HH:MM name: message , with a header recording when it was generated.

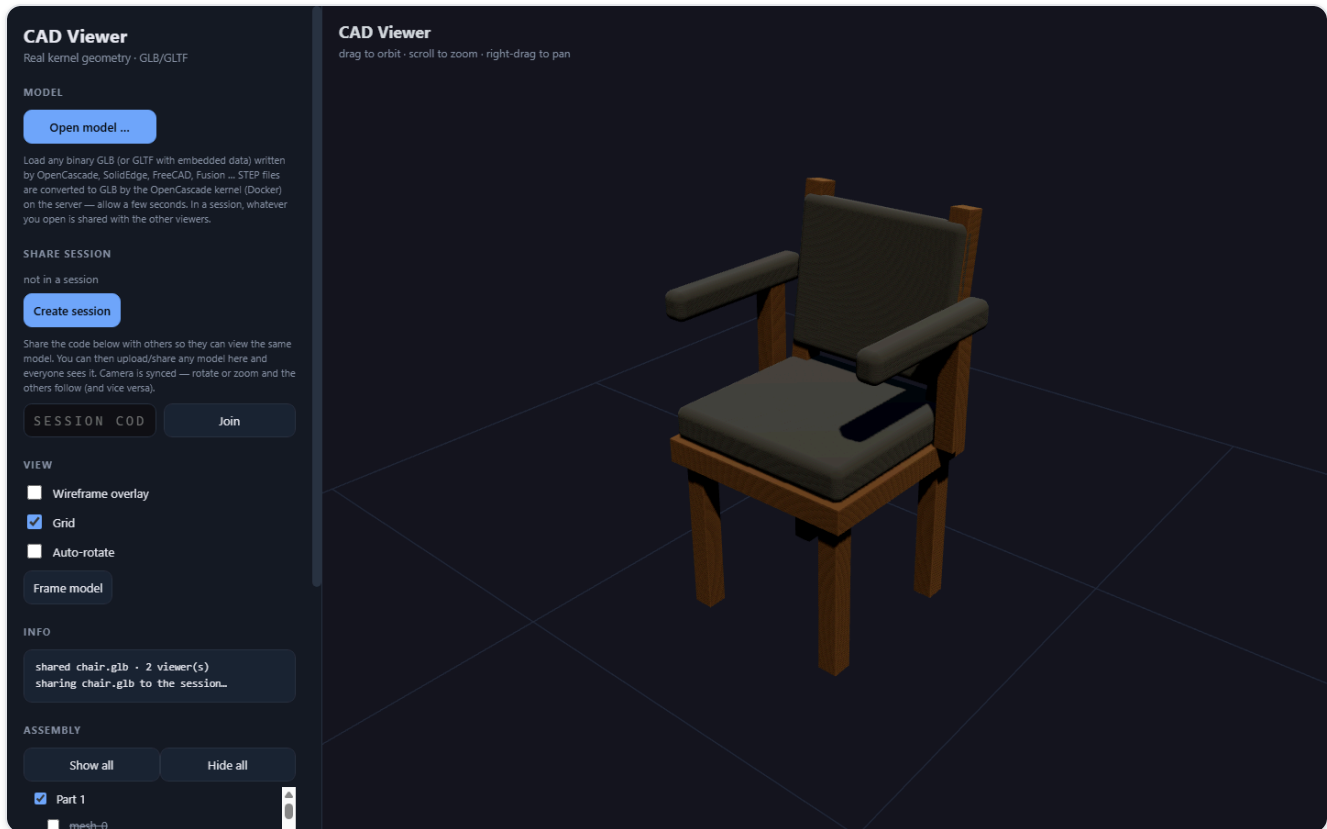
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## 14. Leaving a session

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Click **Leave session** to leave. What happens depends on your role:

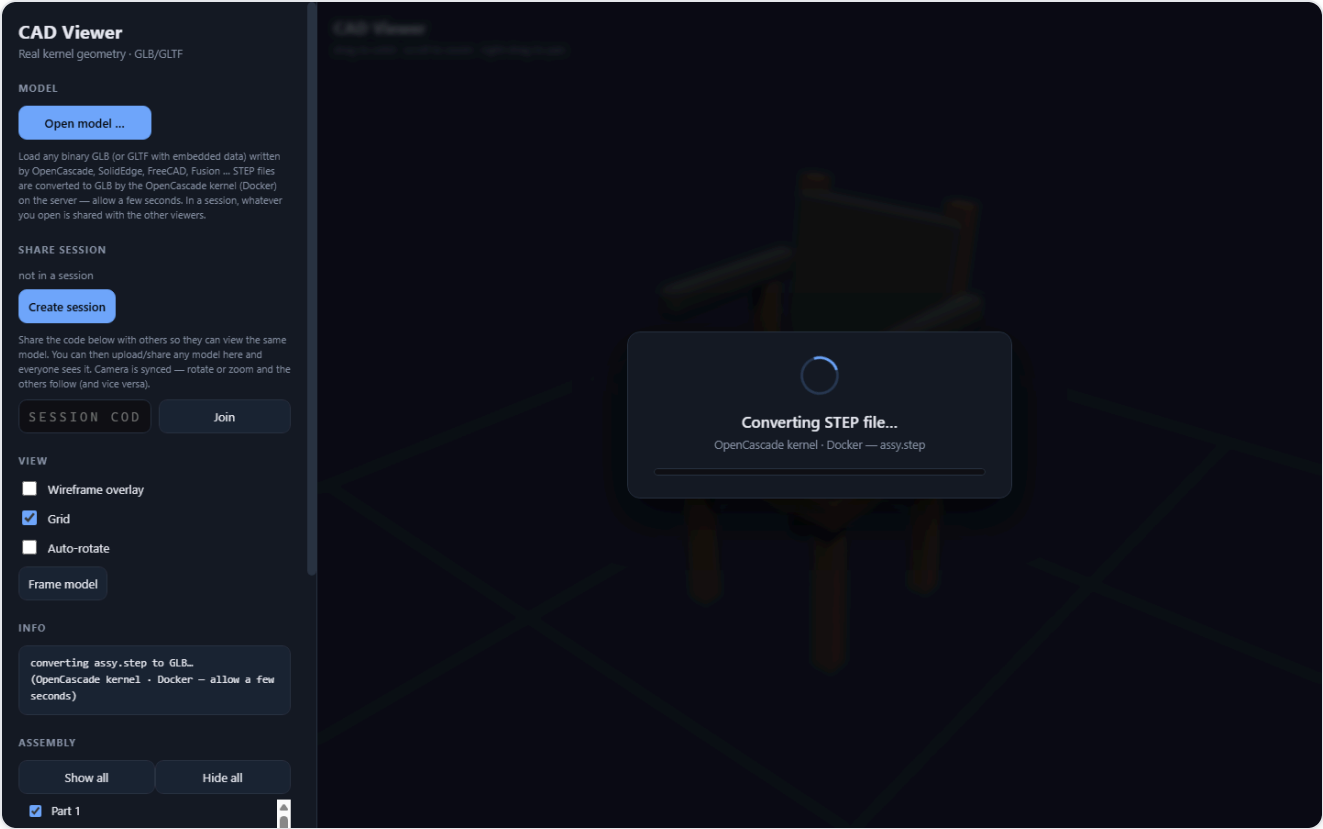
- **Guest** — the shared model is **cleared from your view** when you leave on your own (for security, nothing stays on your screen), and you return to the empty viewer. The session stays alive for the other members.
- **Host** — you keep your local model and return to solo viewing, but the session **ends for everyone else**: the shared model belongs to the host, so when the host leaves (or closes the browser), every guest's shared model is **cleared** and they see "*host left — session ended*". There is no new host — a hostless session is dead.



## 15. Opening a STEP / IGES / STL file

STEP, IGES and STL conversion happens through the OpenCascade kernel (Docker).

The first time you open one you'll see the conversion overlay; when it finishes the GLB is loaded — with the original **colours and materials** preserved, and part names appearing in the Assembly tree.



If Docker or the `chair-cq:local` image isn't installed, these formats show a conversion error while GLB/GLTF continues to work normally.

## 16. Standalone CAD converter (convert-cad)

Besides the viewer, the distribution includes a **separate, standalone CAD converter** — `convert-cad` **(v0.2)** — that turns STEP / IGES / STL files into GLB / GLTF **without** the viewer. It is independent of the viewer's own converters, so using it can never disturb them.

### What it converts

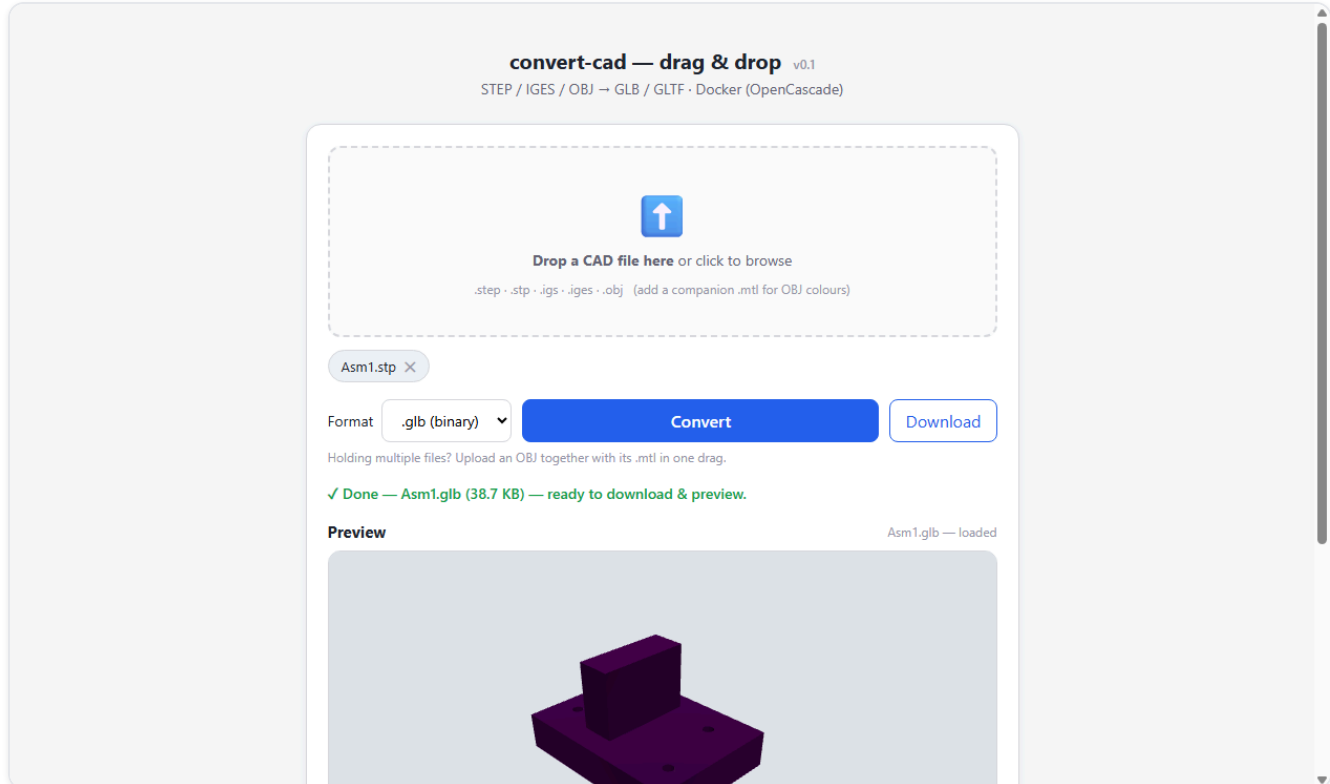
| Input                                  | Notes                                                   |
|----------------------------------------|---------------------------------------------------------|
| <code>.step</code> , <code>.stp</code> | B-rep; keeps assembly part names + per-part colours     |
| <code>.igs</code> , <code>.iges</code> | B-rep; keeps colours; parts named <code>Part1..N</code> |

| Input             | Notes                                                       |
|-------------------|-------------------------------------------------------------|
| <code>.stl</code> | Mesh; converted host-side by the pure-JS writer (no Docker) |

Output format is chosen by the **output file extension**: `.glb` → a single binary file; `.gltf` → text JSON + a companion `.bin`.

## Using the drag-&-drop web UI (recommended)

1. Launch the web UI: double-click `convert-cad-web.bat` (Windows), or run `node convert-cad-server.mjs` from the `tools/convert-cad/` folder.
1. Open `http://localhost:8787/` in a browser.
2. **Drag & drop** a CAD file onto the page (a STEP/IGES/STL file).
3. Pick `.glb` or `.gltf`, click **Convert**.
4. **Preview** the result in the built-in 3D viewer (drag = rotate, scroll = zoom, right-drag = pan), then **Download**.



## Using the command line

```
node convert-cad.mjs input.step output.glb      # or .gltf
```

## Requirements

The converter runs the OpenCascade kernel inside the `chair-cq:local` Docker image — the same one the viewer uses. Docker Desktop must be running and the image built ( `install-docker-opencascade.bat` , or ``docker build -t chair-cq:local .``). **Node.js** is needed for the launcher.

## How it helps sharing

Convert a CAD file to GLB/GLTF once, then hand the lightweight file to someone who opens it directly in the viewer — they never need Docker or the converter. It's the same "one compact file over the network" idea, applied offline.

## 17. Troubleshooting

| Problem                                 | Fix                                                                                                                                |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Can't open the app on another PC        | Use the <b>join link</b> (LAN address), make sure both PCs are on the same network, and that port 8088 isn't blocked by a firewall |
| STEP/IGES/STL shows "conversion failed" | Run <code>install-docker-opencascade.bat</code> ; confirm Docker is running with the <code>chair-cq:local</code> image             |
| Guest doesn't get the model             | The host must be connected with a model loaded — joining an empty session shows nothing until the host shares                      |
| Port already in use                     | Change it: <code>start.bat 4323</code> (zip), a <code>port.txt</code> file, or <code>set PORT=4323</code>                          |
| Join link shows the wrong IP            | The link uses the host's LAN address; refresh/re-create the session to re-detect it                                                |
| Measure tool won't snap                 | Make sure the <b>Measure</b> checkbox is on, and click near a part's <b>corner</b> (the tool snaps to corners, not faces)          |

| Problem                                  | Fix                                                                                                              |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Explode looks like the whole model moves | Set <b>Level</b> to <b>All parts</b> — a single top-level sub-assembly (e.g. one wrapper node) moves as one unit |

## 18. Sharing with external parties

By default the viewer is meant for the **local network (LAN)**. To let someone outside your network view a session, you have two main options.

### Option 1 — Port forwarding on the router

1. Find the server PC's **LAN IP** (e.g. `192.168.x.x` ).
2. In your router's admin page, set up a **port forward**: forward the app's port (default **8088**) to that LAN IP and port.

1. Share `http://<your-public-IP>:8088` with the external party.

**Notes.** The public IP can change (use a dynamic-DNS service if needed).

Opening a port exposes the server to the internet — use this only for trusted people, keep it temporary, and close the forward when you're done. For real deployments, prefer HTTPS (e.g. a reverse proxy like Caddy/Nginx with TLS).

### Option 2 — Deploy on a public / virtual server

1. Provision a small **virtual server** (VPS/cloud instance) with a public IP or domain.
1. Copy the distribution and run the server there ( `node src/server.js` ), or package it into the same Docker image.
1. Share the server's URL with the external party.

This is the best option for **always-on** external access (client demos, vendors/subcontractors joining anytime) because it doesn't depend on a home router or a machine that must stay on.

**Security reminder.** Whatever route you use, anyone who has the link can view the shared model. The viewer streams the model to the guest's memory and writes nothing to disk, but for sensitive or NDA geometry, still share links only with people who are meant to see the model, and consider a private network / VPN (e.g. Tailscale, WireGuard) for the most controlled access.

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*CAD Viewer v0.81 — collaborative CAD viewing for the LAN. · [Changelog](#).*